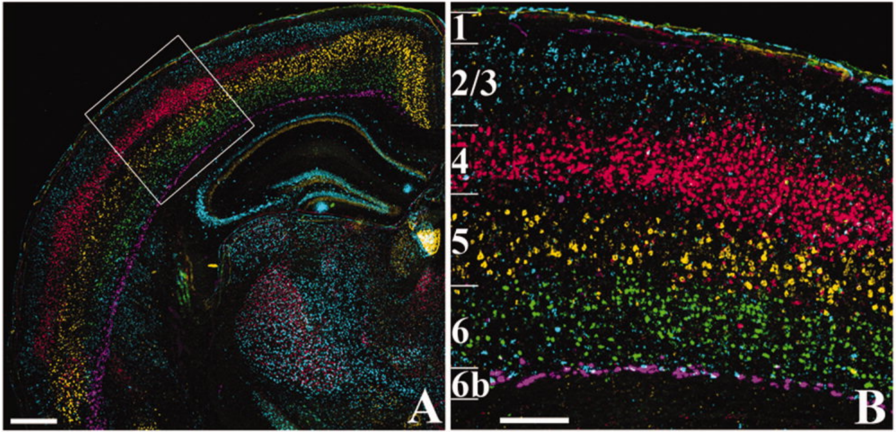


Boyle et al. (2011)

The cortical microcircuit

Layers of cortex

Computational Neuroscience

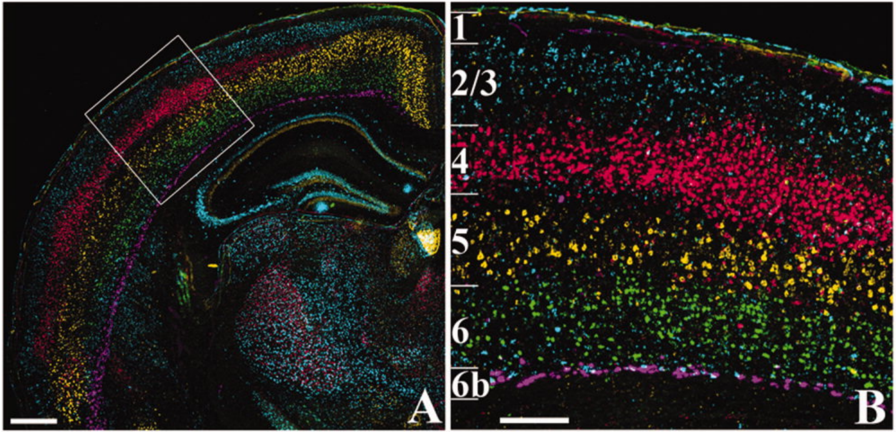
University of Bristol

mrule

Learning outcomes:

- The cortical microcircuit
  - How many layers are (typically) in neocortex?
  - Which is the main layer receiving thalamic input?
  - Which layers give rise to cortico-cortical connections?
  - Makeup of inhibitory, excitatory cell types?

The cortical microcircuit

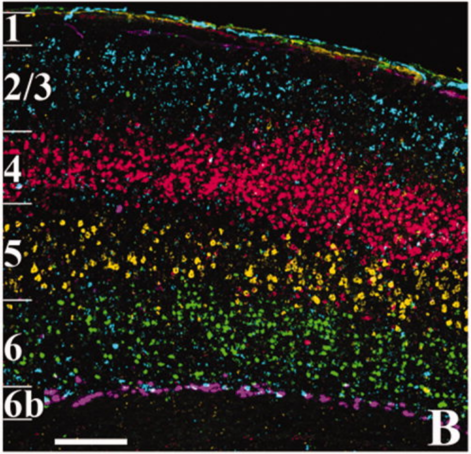


Boyle et al. (2011)

~~Onions have~~ Cortex has *layers*

6-layer neocortex

1. (top, mostly neurites)
2. Gets cortico-cortical) **V1 complex cells**
3. Sends cortico-cortical) **V1 complex cells**
4. Gets thalamic input, e.g. sensory) **V1 simple cells**
5. sends signals out of the brain, e.g. motor
6. (deepest) feedback to thalamus



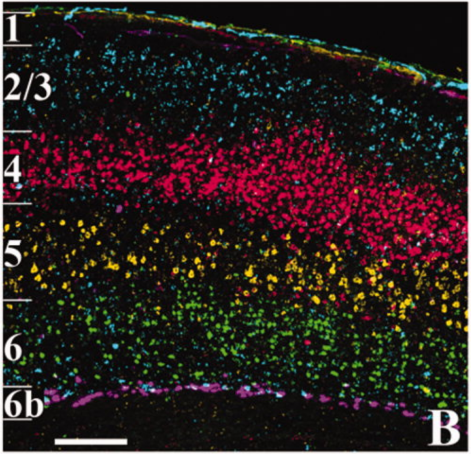
Boyle et al. (2011)

Only visual tuning cell-types ↔ layers on 2025 final, hippocampus on there but not about layers

~~Onions have~~ Cortex has *layers*

Parts of the brain that evolved earlier have fewer layers

- 3-layer archicortex (123)
  - Hippocampus: Dentate Gyrus, CA3, CA1
- 3–4-layer paleocortex (not covered) (1235)
- 3–4-layer periarchicortex (1235)
  - Subiculum
- 5–6-layer mesocortex (123456)
  - Entorhinal Cortex



Boyle et al. (2011)

Only visual tuning cell-types ↔ layers on 2025 final, hippocampus on there but not about layers

The cortical microcircuit

The mammalian neocortex is fairly homogeneous: most parts of it look similar (with important exceptions). Some aspects of V1 architecture apply to cortex in general

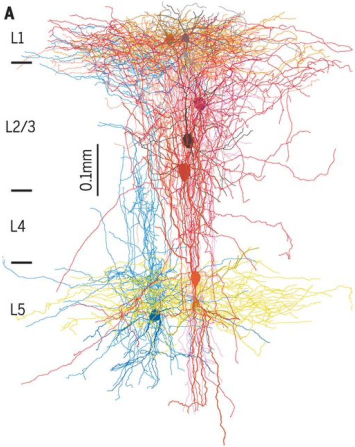
Each layer has a mix of distinct excitatory and inhibitory cell types

- More excitatory neurons than inhibitory neurons
- $\approx 80/20\%$ E/I split
- More types of inhibitory (10–20) than excitatory (4–6)

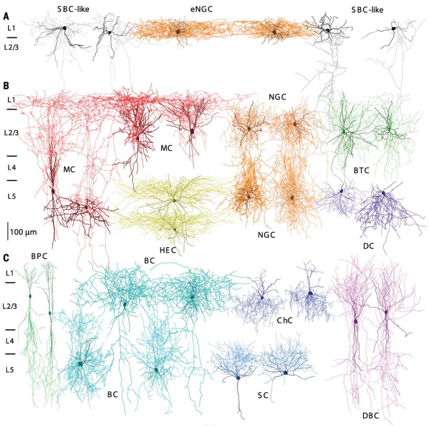
Information flow:

- Thalamus $\rightarrow$ layer 4 $\rightarrow$ layer 2/3 $\rightarrow$ other parts of the brain and layer 5
- Layer 5: outputs (usually motor-like)
- Layer 6: Feedback modulation of cortico-thalamic loops

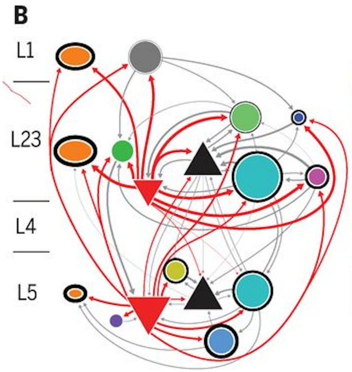
Excitatory (glutamatergic) pyramidal cells



Inhibitory (GABAergic) interneurons



Circuit Diagram



Jiang et al. (2015)

pause

6-layered neocortex (oversimplified)

- Layer 4: ____ from thalamus, e.g. retinal signals arrive in V1 via LGN
- Layer 2/3: Local recurrent processing, sends signals within _____
- Layer 5: Sends motor-like or motor-related outputs
- Layer 6: Closes thalamic feedback loop

V1 tuning curves

- Simple cells
  - Found in ______________ that receive inputs from the LGN (part of the thalamus)

- Complex cells
  - Found in ____

6-layered neocortex (oversimplified)

- Layer 4: Input from thalamus, e.g. retinal signals arrive in V1 via LGN
- Layer 2/3: Local recurrent processing, sends signals within cortex
- Layer 5: Sends motor-like or motor-related outputs
- Layer 6: Closes thalamic feedback loop

V1 tuning curves

- Simple cells
  - Found in V1 layers 4 and 6 that receive inputs from the LGN (part of the thalamus)

- Complex cells
  - Found in V1 layers 2/3 and 6, and higher-order visual areas

end

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